CDS Demo Capstone – Data Type Table (UML Class Diagram)

SAP BTP ABAP Environment | 1 table, 41 columns

«table»

zvkrap01_dt_demo

Keys and numbers (12 cols)		Characters (9 cols)		Numeric text (5 cols)		Date and time (9 cols)		Special types (6 cols)	
PK client	abap.clnt	t_char1	char(1)	t_numc4	numc(4)	t_dats	abap.dats	t_lang	abap.lang
PK rec_id	abap.int4	t_char3	char(3)	t_numc6	numc(6)	t_datn	abap.datn	t_cuky	abap.cuky
case_desc	char(60)	t_char5	char(5)	t_numc8	numc(8)	t_tims	abap.tims	t_curr	curr(9,2)
t_int1	abap.int1	t_char6	char(6)	t_numc10	numc(10)	t_timn	abap.timn	t_unit	unit(3)
t_int2	abap.int2	t_char8	char(8)	t_numc14	numc(14)	t_utc	abap.utclong	t_quan	quan(9,3)
t_int4	abap.int4	t_char10	char(10)			t_tstmp	dec(15,0)	t_raw	raw(8)
t_int8	abap.int8	t_char30	char(30)			t_tstmpl	dec(21,7)		
t_dec	dec(15,3)	t_sstr	sstring(60)			t_tzone	char(6)		
t_dec2	dec(9,2)	t_str	string(0)			t_accp	abap.accp		
t_df16	decfloat16								
t_df34	decfloat34								
t_fltp	abap.fltp								

Columns that belong together (same colored dot)

- **Date cast pair** — t_char8 · t_numc8
- **Time cast pair** — t_char6 · t_numc6
- **Number cast pair** — t_char10 · t_numc10
- **Currency reference** — t_cuky · t_curr
- **Unit reference** — t_unit · t_quan
- **Date generations** — t_dats · t_datn
- **Time generations** — t_tims · t_timn
- **PK Primary key** — client, rec_id

CDS Demo – Scenarios: String Functions and Expressions

Group	ID	Scenario	CDS feature shown	Columns used	Rows must show	Source
String functions and expressions	01	Trim blanks	ltrim(t_char30, ' '), rtrim(t_char30, ' ')	t_char30, t_sstr	Leading, trailing, both, none	SAP
	02	Trim a given character	ltrim(t_char10, '0'), rtrim(t_char10, 'X')	t_char10	Leading zeros, trailing X	SAP
	03	Pad to a length	lpad(t_char10, 12, '#'), rpad(...)	t_char10, t_numc4	Shorter and longer than the length	SAP
	04	Upper and lower case	upper(t_char30), lower(t_char30)	t_char30	Mixed case, one with umlauts	SAP
	05	Combine text, numbers and codes	concat(a, b), concat_with_space(a, b, 1)	t_char30, t_numc6, t_sstr	Character, numc and sstring together	SAP
	06	Cut out part of a text	left, right, substring(t_char30, 4, 3)	t_char30	Positions inside the text	SAP
	07	Find and measure	instr(t_char30, '@'), length(t_char30)	t_char30	Text with and without @	SAP
	08	Replace	replace(t_char30, '-', '')	t_char30	Spaces, hyphens, plus sign, brackets	SAP
	09	Replace by pattern	replace_regexpr(PCRE => '\d', VALUE => t_sstr, WITH => '#')	t_sstr	Letters, digits and symbols mixed	SAP
	10	string versus sstring	Restrictions of string in CDS	t_str, t_sstr	One long text	Check

Source: SAP = function and syntax found in SAP documentation or SAP's own sample. Check = not confirmed, test in ADT.

CDS Demo – Scenarios: Arithmetic and Mathematical

Group	ID	Scenario	CDS feature shown	Columns used	Rows must show	Source
Arithmetic expressions	11	Integer arithmetic	+ - *	t_int1, t_int2, t_int4, t_int8	Positive and negative values	SAP
	12	Mixed types	int and dec in one expression	t_int4, t_dec, t_dec2	Result type and precision	Check
	13	Division with the / operator	/	t_dec, t_dec2	Results with repeating decimals	SAP
	14	Percentage of a value	cast(a / b * 100 as abap.dec(10,2))	t_dec, t_dec2	Percentages 0 to 100	SAP
	15	Amount times quantity	Arithmetic after cast to dec	t_curr, t_quan	Typical amounts and quantities	Check
	16	Amount as decfloat	curr_to_decfloat_amount, get_numeric_value	t_curr, t_cuky	Amounts of different size	Check
Mathematical: float and decimal precision	17	Float to decimal	fltp_to_dec(t_fltp as abap.dec(15,3))	t_fltp	Large and small values	SAP
	18	decfloat arithmetic	decfloat16 against decfloat34	t_df16, t_df34	Values with many digits	Check
	19	The same value in three types	Compare dec, decfloat and fltp	t_dec, t_df34, t_fltp	0.1 and 1/3	Check

Source: SAP = function and syntax found in SAP documentation or SAP's own sample. Check = not confirmed, test in ADT.

CDS Demo – Scenarios: Numeric Functions and Conditions

Group	ID	Scenario	CDS feature shown	Columns used	Rows must show	Source
Numeric functions	20	Absolute value	abs(t_int4)	t_int4, t_dec	Positive and negative	SAP
	21	Round up and down	ceil(t_df34), floor(t_df34)	t_df34, t_dec	Values at .5, negative values	SAP
	22	Round to decimals	round(t_df34, 2)	t_df34, t_dec	Halves, negative values	SAP
	23	Integer division and remainder	div(t_int4, t_int2), mod(...)	t_int4, t_int2	Negative values, remainder zero	SAP
	24	Division with decimals	division(t_dec, t_dec2, 2)	t_dec, t_dec2	Different numbers of decimals	SAP
Conditions	25	Simple CASE	case t_char1 when 'A' then ... else ... end	t_char1	Every branch and the ELSE	SAP
	26	Searched CASE	case when t_int4 <= 10 then ... end	t_int4, t_dec	Every range and the ELSE	SAP
	27	CASE without ELSE	No ELSE returns the null value; ELSE NULL	t_char1	A value that matches no branch	SAP
	28	Filter by pattern and range	BETWEEN, LIKE, IS INITIAL, IS NULL	t_int4, t_char30, t_dats	Common prefixes, initial values	SAP

Source: SAP = function and syntax found in SAP documentation or SAP's own sample. Check = not confirmed, test in ADT.

CDS Demo – Scenarios: Date and Time

Group	ID	Scenario	CDS feature shown	Columns used	Rows must show	Source
Date and time	29	Validity of a date	dats_is_valid(t_dats)	t_dats	00010101 valid, 00000000 invalid, initial	SAP
	30	Validity of a time	tims_is_valid(t_tims)	t_tims	Valid, impossible, initial	Check
	31	Days between two dates	dats_days_between(t_dats, $session.system_date)	t_dats	Later, earlier, same day, invalid date	SAP
	32	Add days and months	dats_add_days(t_dats, 7, 'INITIAL'), dats_add_months(...)	t_dats	31 January, leap day, year end	SAP
	33	Days between on DATN	datn_days_between(dats_to_datn(...), ...)	t_dats, t_datn	Same cases as above	SAP
	34	Add days on DATN	datn_add_days(t_datn, 7, ...)	t_datn	Same cases as above	Check
	35	Timestamp to date and time	tstmp_to_dats(t_tstmp, t_tzone, 'NULL'), tstmp_to_tims	t_tstmp, t_tzone	Different time zones	SAP
	36	Seconds between, add seconds	tstmp_seconds_between, tstmp_add_seconds	t_tstmp	Two moments apart	Check
	37	UTC long: now and add seconds	utcl_current( ), utcl_add_seconds(t_utc, 60)	t_utc	A value and the value a minute later	SAP
	38	UTC long: seconds between	utcl_seconds_between(t_utc, ...)	t_utc	Two moments apart	Check
	39	Combine date and time	dats_tims_to_tstmp(t_dats, t_tims, ...)	t_dats, t_tims, t_tzone	Same date, different zones	Check
	40	Posting period from a date	cast dats to char, left, cast to accp	t_dats, t_accp	Month end, year end	Check

Source: SAP = function and syntax found in SAP documentation or SAP's own sample. Check = not confirmed, test in ADT.

CDS Demo – Scenarios: Conversion, Aggregates, Literals

Group	ID	Scenario	CDS feature shown	Columns used	Rows must show	Source
Conversion	41	Quantity in another unit	unit_conversion(quantity =>, source_unit =>, target_unit =>)	t_quan, t_unit	g, kg and one unit of another kind	Check
	42	Amount in another currency	currency_conversion(amount =>, source_currency =>, target_currency =>, exchange_rate_date =>)	t_curr, t_cuky, t_dats	EUR, USD, JPY; rates must exist	Check
Aggregates	43	Totals per group	sum(t_curr) with GROUP BY	t_char5, t_curr, t_cuky	Several rows per group, one currency	SAP
	44	Statistics per group	avg(t_dec as abap.dec(15,2)), min, max, count(*)	t_char5, t_dec, t_dats	Groups of different size	SAP
	45	Distinct values and HAVING	count(distinct t_char3), HAVING	t_char5, t_char3	Repeated sub-groups	SAP
	46	Aggregate over numeric text	sum(cast(t_numc4 as abap.int4))	t_char5, t_numc4	Leading zeros, one large value	SAP
Literals, parameters, session	47	Literals, parameters, session variables	abap.int4'12345', $parameters.p, $session.system_date	none	Any row	SAP

Source: SAP = function and syntax found in SAP documentation or SAP's own sample. Check = not confirmed, test in ADT.

CDS Demo – Scenarios: Casting (1 of 2)

Group	ID	Scenario	CDS feature shown	Columns used	Rows must show	Source
Casting: one of each matrix code	48	Integer to a larger integer	[x] cast int to int8	t_int4	Negative, zero, large value	SAP
	49	Decimal to decimal and float	[x] cast dec to dec, fltp	t_dec	Many decimals, negative values	SAP
	50	Numeric text to a number	[x] cast numc to dec, fltp	t_numc14	14-digit timestamp, all zeros	SAP
	51	Text to shorter text and sstring	[x] cast char to char, sstring	t_char30	Text longer than the target	SAP
	52	Text to date	[y] cast char to dats	t_char8	Valid, impossible, blank	SAP
	53	Text to time	[y] cast char to tims	t_char6	Valid, impossible, blank	SAP
	54	Text to numeric text	[y] cast char to numc	t_char10	Digits, digits with blanks, letters	SAP
	55	Text to language, unit, currency	[y] cast char to lang, unit, cuky	t_char1, t_char3, t_char5	E, KG, EUR	SAP
	56	Numbers to text	[y] cast int, dec to char	t_int4, t_dec	Negative, zero, many decimals	SAP
	57	Numeric text to integers	[y] cast numc to int, int8	t_numc10	Leading zeros, one large value	SAP
	58	Date and time to text	[y] cast dats, tims to char	t_dats, t_tims	Normal and initial values	SAP

Source: SAP = function and syntax found in SAP documentation or SAP's own sample. Check = not confirmed, test in ADT. x, y, z, d, p and - are the cell codes of the cast matrix in the ABAP keyword documentation (7.51). The page I could read does not explain them, so check the legend there.

CDS Demo – Scenarios: Casting (2 of 2)

Group	ID	Scenario	CDS feature shown	Columns used	Rows must show	Source
Casting: one of each matrix code	59	Numeric text to date	[z] cast numc to dats	t_numc8	Valid, 00000000, impossible	SAP
	60	Numeric text to time	[z] cast numc to tims	t_numc6	Valid, 000000, impossible	SAP
	61	Numeric text to shorter numeric text	[z] cast numc to numc	t_numc10	Value longer than the target	SAP
	62	Posting period to text	[z] cast accp to char, numc	t_accp	202610, initial	SAP
	63	Language, unit, currency to text	[d] cast lang, unit, cuky to char	t_lang, t_unit, t_cuky	E, KG, EUR	SAP
	64	Text to posting period	[p] cast char to accp	t_char6	202610, 000000, invalid month	SAP
	65	Numeric text to posting period	[p] cast numc to accp	t_numc6	202610, 000000, invalid month	SAP
	66	Same type to same type	[p] cast raw to raw, lang to lang	t_raw, t_lang	Any value	SAP
	67	Combinations marked [-]	cast char to dec, int to numc	t_char10, t_int4	Syntax error shown in ADT	SAP

Source: SAP = function and syntax found in SAP documentation or SAP's own sample. Check = not confirmed, test in ADT. x, y, z, d, p and - are the cell codes of the cast matrix in the ABAP keyword documentation (7.51). The page I could read does not explain them, so check the legend there.

CDS Demo – Scenarios: Set Operations and Relations

Group	ID	Scenario	CDS feature shown	Columns used	Rows must show	Source
Set operations and relations	68	UNION ALL of two selections	UNION ALL	t_char5, t_char1	Rows that match both selections	Check
	69	UNION of two selections	UNION	t_char5, t_char1	Same rows, duplicates removed	SAP
	70	Inner self join	INNER JOIN of the table with itself	rec_id, t_char3	Rows paired by the same code	SAP
	71	Left outer self join with coalesce	LEFT OUTER JOIN, coalesce(..., 'N')	rec_id, t_int4	Rows with and without a match	SAP
	72	Right outer and cross join	RIGHT OUTER JOIN, CROSS JOIN	rec_id, t_char3	Small sets of rows	SAP
	73	Association to the table itself	association [0..1] to ... as _ref	rec_id, t_int4	Rows that point to an existing row	SAP
	74	Rows of one group only	Input parameter, $parameters.p	t_char5	At least three groups	SAP

Source: SAP = function and syntax found in SAP documentation or SAP's own sample. Check = not confirmed, test in ADT.

CDS Demo Capstone – What Stays on the Existing Capstone Tables

Joins, associations, order totals

Topic	Existing capstone tables	Example
Inner join	Sales Order Header, Customer	Orders with the customer name
Left outer join	Customer, Sales Order Header	Customers without orders
Right outer join, cross join	Product, Customer, Sales Order Item	Any two small sets of rows
Associations and path expressions	Header, Customer, Item, Product	Buyer name on an order, product on an item
Composition and to-parent	Sales Order Header, Sales Order Item	Already modelled in the capstone
Aggregates over items	Sales Order Item	Order total, count of items
Value help, access control	Customer	Value help for the buyer; restrict by country
UNION and UNION ALL	Data type table (with itself)	The two union scenarios

⚠️ **Check the data:** left outer join needs at least one customer without an order, and the aggregates need orders with several items. If the capstone data does not have them, add rows. The data type table has no parent column, so its self joins use rec_id against t_int4.

CDS Demo – What Was Validated Against SAP Sources

Validation | checked 11 Oct 2026

Area	SAP source used	Result
String functions	ABAP keyword documentation 7.52, SQL functions: string functions; SAP ABAP cheat sheet sample (view entity)	All 14 functions confirmed. replace_regexpr confirmed in the sample only
Numeric functions	SAP ABAP cheat sheet sample (view entity)	abs, ceil, floor, div, mod, division, round confirmed with their arguments
Date functions	ABAP keyword documentation 7.52, CDS date functions; cheat sheet sample	dats_is_valid, dats_days_between, dats_add_days, dats_add_months, datn_days_between, dats_to_datn confirmed
Timestamp functions	ABAP keyword documentation, ABAP SQL timestamp functions; SAP Community blog, release 7.51	utcl_current and utcl_add_seconds confirmed in CDS. tstmp_* confirmed for ABAP SQL, tstmp_to_dats/tims in CDS
Conversion, aggregates, joins	SAP ABAP cheat sheet samples (view entity)	unit_conversion, fltp_to_dec, coalesce, CASE, aggregates, joins, associations confirmed
Cast matrix	ABAP keyword documentation 7.51 (CDS) and 7.53 (ABAP SQL)	Matrix confirmed. Legend for x, y, z, d, p not retrievable. The two releases differ

⚠️ **Blocked:** many pages under help.sap.com could not be opened by my tools. Anything that depends on them is marked Check.

CDS Demo Capstone – Still To Confirm in ADT

Corrections made	ltrim and rtrim need a character argument and also remove trailing blanks. && removed. dats_add_days and dats_add_months need an on_error literal. fltp_to_dec needs a target type. unit_conversion needs source_unit. round and division need a decimals argument.
Cast matrix	Legend of x, y, z, d, p not retrieved. The 7.53 ABAP SQL matrix shows DEC to INT as allowed. The 7.51 CDS matrix does not. Test every cast on your system.
Functions marked Check	tims_is_valid, datn_add_days, tstmp_seconds_between and tstmp_add_seconds in CDS, utcl_seconds_between in CDS, dats_tims_to_tstmp signature, currency_conversion syntax, curr_to_decfloat_amount, get_numeric_value, UNION ALL, string restrictions.
Operand types	Which column types each function accepts (for example concat on numc, ceil on dec) and the behaviour for negative values, division by zero and NULL are not confirmed.
Column types	int1, int2, int8, decfloat16/34, fltp, lang, quan, unit, raw and accp are not in SAP's ABAP Cloud cheat sheet. Confirm that the table activates.
Size	74 scenarios on one table is a proposal. Drop any scenario you do not teach, then drop the columns only that scenario uses.